

Regulated Document Decisioning

A Governance and Operating-Model Memo

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Executive Summary

Over a multi-week build program on the UiPath platform, I delivered three increasingly sophisticated automation systems for a regulated financial-services context. The program advanced deliberately: from deterministic rule-based automation, to agentic LLM-driven processing, to a synthesis of both - **Regulated Document Decisioning (RDD)**, an end-to-end pipeline that ingests a loan application or an insurance claim, decisions it, subjects adverse outcomes to human review, verifies regulatory compliance, generates a statutory customer letter, archives that letter to an immutable audit store, and notifies the customer.

RDD is the capstone, and it runs end to end today. Its defining characteristic is not that it automates a decision - it is that it produces, for every decision, a complete and examiner-ready governance record: who decided, on what basis, under which regulatory frameworks, with what human involvement, and with what was communicated to the customer. The thesis of the program is that agentic automation in a bank is viable only if governance is a first-class output rather than an afterthought. RDD is built to demonstrate exactly that.

This memo summarizes the three-build progression, describes RDD's architecture and governance model, lays out its operating model, and assesses its current maturity and roadmap.

The Problem

Regulated document decisioning is one of the most expensive and highest-risk routine workloads a bank carries. Every loan application and every insurance claim that ends in an adverse outcome triggers a statutory obligation: a specific, reasoned, time-bound written notice to the customer. Reg B (the Equal Credit Opportunity Act) requires adverse-action notices on credit decisions with specific - not vague - reasons. ERISA Section 503 requires written denial notices on benefit claims with a defined appeal window. Behind both sits the examiner: every one of these decisions must be reconstructable on demand, years later, with its full basis intact.

Today this work is largely manual, and manual handling fails in three predictable ways. It is slow and inconsistent - a small consumer loan waits in the same queue as a complex commercial request. It is error-prone - adverse-action notices with insufficiently specific

reasons are themselves a compliance violation. And it is poorly documented - the reasoning behind a decision lives in an analyst's head or a scattered email thread, not in a structured, queryable record. The cost is real, but the risk is larger than the cost.

The program set out to answer a focused question: **can automation absorb this workload while making the governance posture stronger, not weaker?**

The Three-Build Progression

The program was structured as three builds, each answering a distinct question and feeding the next.

Build 1 - Loan Application Intake (deterministic RPA). The first build asked whether rule-based automation could absorb the routine majority of an intake workload while preserving human judgment for material decisions. It was implemented on classic UiPath with the dispatcher/performer pattern: a Loader process ingesting applications from a shared inbox and writing durable queue items, and a Worker process built on the Robotic Enterprise Framework consuming them, enriching each with customer and credit data, and routing to one of five outcomes - auto-approve, auto-decline, escalate for underwriting review, manual review, or business exception. It established the disciplines the rest of the program would rely on: clean separation of concerns, externalized and configurable policy thresholds, a rigorous three-class exception taxonomy, versioned reusable libraries, and a human handoff through Action Center for high-value cases. It was load-tested at 500 applications and scaled cleanly. Build 1 proved that the predictable cases can be handled with policy precision, leaving humans the decisions where their judgment actually changes the outcome.

Build 2 - Cascade Health Claims Adjudication (agentic). The second build asked whether agentic, LLM-driven systems could do the same for work requiring genuine reasoning rather than rule evaluation. It introduced two autonomous LLM agents - one extracting structured claim data from inbound PDFs, one adjudicating under five hard policy rules with access to an independent "second opinion" tool - orchestrated through UiPath Maestro, with a suspend-and-wait human-review model that keeps a process instance alive for days while a reviewer responds. Build 2 proved agentic orchestration works. More importantly, it produced a clear-eyed catalog of the *governance hazards* agentic systems introduce: contract drift between an agent's declared inputs and what its tools actually produce; value-domain mismatches between prompt-defined outputs and orchestrator routing logic that schema validation cannot catch; and the silent operational risk of external credential lifecycles. The central lesson of Build 2 was that the hard problems of agentic automation are governance problems, not wiring problems.

Build 3 - Regulated Document Decisioning (the synthesis). The capstone combines both. Where policy is unambiguous - loan decisioning against credit-score floors, loan-to-income ceilings, and amount thresholds - RDD uses a deterministic rule ladder inherited directly from Build 1. Where the work requires reasoning - adjudicating a medical claim's necessity and policy alignment - it uses an LLM agent inherited from Build 2. And it adds

what neither predecessor had: a dedicated regulatory compliance layer and a customer disclosure layer that together turn a raw decision into a compliant, documented, and communicated outcome. RDD is where deterministic precision, agentic reasoning, and regulatory governance meet in a single pipeline.

What RDD Is: Architecture

RDD is a UiPath Studio Web Maestro BPMN process. An inbound application or claim arrives by email and moves through a five-stage agent pipeline.

1. **Document Classification.** An LLM agent classifies the submission as a loan application, an insurance claim, or unknown - routing unknowns to human review rather than guessing.
2. **Intake.** A branch-specific intake step validates and structures the payload, computes derived fields (loan-to-income and payment-to-income ratios, claim timeliness, amount tiers), and flags data-quality issues.
3. **Risk Decisioning.** Loans run through a deterministic credit-decisioning ladder; claims run through an LLM adjudication agent applying medical-necessity and policy-based reasoning. Both converge on a disposition.
4. **Regulatory Compliance.** An LLM compliance agent verifies the disposition against every applicable framework - Reg B/ECOA, FCRA, TILA for credit; ERISA Section 503, state insurance codes, HIPAA for claims; OFAC sanctions screening and SR 11-7 model-risk standards across both - and produces a structured, evidence-citing compliance assessment.
5. **Customer Disclosure.** A final LLM agent drafts the customer communication: a short cover email and a formal statutory letter, written to the specificity that Reg B and ERISA require.

Between risk decisioning and compliance sits a **human-review gate**. Any case the system flags - borderline, large, data-flagged, or escalation-required - suspends for a human reviewer through a dedicated Action Center app; the reviewer's decision and rationale are captured back into the record. Clean, unambiguous cases proceed automatically with their disposition set deterministically.

After disclosure, the pipeline generates the letter as a PDF, archives it to an internal Storage Bucket under a content-addressed path, updates the decision record with the archival metadata, retrieves the archived file, and emails it to the customer. The inbound email is marked processed so the same submission cannot be decisioned twice.

The Governance Model

RDD's governance model is the point of the project, not a feature of it.

An immutable decision record. Every run carries a single decisionEvent object that accumulates the full chain - classification, intake, risk output, compliance findings, the drafted communication, and every human touchpoint - under one stable decision identifier minted at intake. That identifier flows through the entire pipeline and becomes the reference number on the customer's letter. One decision, one record, one identifier, reconstructable end to end.

Framework coverage as an explicit output. The compliance agent does not merely "check compliance" - it names the governing frameworks for each decision and records a discrete, evidence-citing finding against each. Every finding must answer the examiner's implicit question - *how do you know?* - from its own text, citing specific values from the case rather than generic assurances.

Human-override scrutiny. Under SR 11-7, when a human overrides the model's recommendation, that override is flagged, the reviewer's rationale is required and retained, and the compliance layer raises a finding specifically on it. Human judgment is preserved - and it is audited.

Statutory customer communication. Adverse outcomes produce notices written to regulatory specificity: Reg B's prohibition on vague rationale, ERISA 503's 180-day appeal window, the required disclosures attached. The communication is treated as a statutory artifact, not a courtesy.

Audit archival. Every letter is archived to an immutable store at the moment of generation, addressed by the decision's reference number, with retention sized to the longest applicable regulatory horizon.

The Operating Model

RDD draws a clear line between where the machine acts and where humans sit. The machine handles classification, intake structuring, deterministic decisioning, compliance verification, and communication drafting - the high-volume, policy-bound work. Humans sit at exactly one place: the review gate, as reviewers of escalated cases and as the final authority whose decision the system records and defers to.

The escalation ladder is deliberate. Hard knockouts - decisions where policy is unambiguous - are decided automatically; routing a hopeless application to a senior reviewer wastes the scarce resource. Borderline, large-exposure, or data-flagged cases escalate. Unknown document types escalate rather than risk misrouting. This mirrors how a real institution allocates reviewer attention: to the cases where judgment changes the outcome.

Intake integrity is enforced at the boundary - inbound submissions are marked processed once claimed, so a retrIGGERED or duplicated email cannot produce a second decision. Retention is set to span the longest regulatory obligation across Reg B and ERISA, so the audit trail outlives any examination window.

Maturity and Roadmap

What runs today. The full pipeline is operational end to end: a real inbound email produces a classified, decisioned, compliance-verified, human-reviewed-where-required, archived, and delivered outcome with a complete decision record.

What was hardened. A focused hardening pass closed the gaps between "works" and "regulator-grade": stable decision identifiers flowing intact through every stage, deterministic disposition-setting on the automated path, consistent institutional identity across all customer-facing artifacts, customer-identity capture at intake, and duplicate-submission protection.

Near-term hardening. The remaining backlog is non-blocking and roadmapped: extracting escalation logic into a fully deterministic step, unifying decision vocabularies across the loan and claim branches, content-hashing archived letters for tamper-evidence, formalizing the retention policy, and resolving templated placeholders against a branch-context lookup.

Phase 2 - making the record queryable. The next phase moves the decision record into UiPath Data Fabric as a first-class entity and builds an operations dashboard over it, so compliance and operations staff can filter decisions by type, disposition, and framework, and drill directly into any archived letter. It also replaces the inline sanctions-screening mock with a real watchlist service and adds a regulatory-citation service the agents can call when constructing their reasoning.

Phase 3 - the governance packet. The final phase produces the formal governance documentation a bank's model-risk function would require: a model card, a decision-logging specification, an audit-retention matrix, and a human-override audit-trail standard.

Recommendation

RDD demonstrates that agentic automation can meet a bank's governance bar - that the same pipeline which removes routine cost can simultaneously strengthen the audit posture, because every decision now produces a structured, examiner-ready record that manual handling never reliably produced. The three-build progression shows the capability was earned deliberately: deterministic automation first, agentic reasoning second, and the regulatory synthesis third.

I recommend proceeding to Phase 2 to make the decision record queryable and the system observable, and to Phase 3 to formalize the governance documentation. RDD is, in its current state, a complete and demonstrable proof of the program's thesis: in a regulated institution, governance is not the constraint on automation - it is the product of doing automation well.